TUGAS 4

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PROGRAM STUDI TEKNIK INFORMATIKA

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Tugas 1: Eksperimen dengan Data Berbeda

Jalankan setiap algoritma dengan data berikut dan catat hasilnya:

Data Set 1:

P1: AT=0, BT=10

P2: AT=1, BT=5

P3: AT=2, BT=8

Data Set 2:

P1: AT=0, BT=5

P2: AT=1, BT=3

P3: AT=2, BT=8

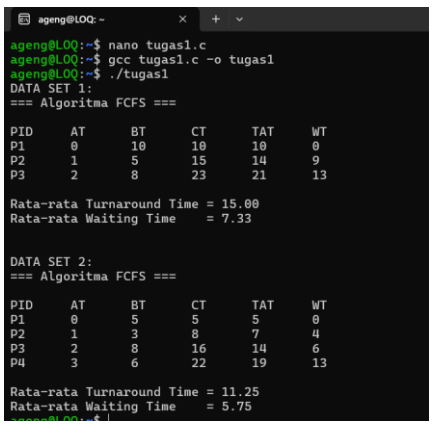
P4: AT=3, BT=6

CODE

```
#include <stdio.h>

struct Process {
    int pid;
    int at;  // Arrival Time
    int bt;  // Burst Time
    int ct;  // Completion Time
    int tat; // Turnaround Time
    int wt;  // Waiting Time
};

// Fungsi untuk menampilkan hasil
void printResult(struct Process p[], int n) {
    float avg_tat = 0, avg_wt = 0;
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n", p[i].pid, p[i].at, p[i].bt, p[i].ct, p[i].tat, p[i].wt);
```

```
        avg_tat += p[i].tat;
        avg_wt += p[i].wt;
    }
    printf("\nRata-rata Turnaround Time = %.2f", avg_tat / n);
    printf("\nRata-rata Waiting Time    = %.2f\n", avg_wt / n);
}

// Algoritma FCFS (First Come First Serve)
void FCFS(struct Process p[], int n) {
    printf("\n=== Algoritma FCFS ===\n");
    int time = 0;
    for (int i = 0; i < n; i++) {
        if (time < p[i].at) time = p[i].at;  // tunggu jika proses belum datang
        time += p[i].bt;
        p[i].ct = time;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt  = p[i].tat - p[i].bt;
    }
    printResult(p, n);
}

int main() {
    // ===== Dataset 1 =====
    struct Process data1[] = {
        {1, 0, 10},
        {2, 1, 5},
        {3, 2, 8}
    };
    int n1 = 3;
```

```
    // ===== Dataset 2 =====
    struct Process data2[] = {
        {1, 0, 5},
        {2, 1, 3},
        {3, 2, 8},
        {4, 3, 6}
    };
    int n2 = 4;

    printf("DATA SET 1:");
    FCFS(data1, n1);

    printf("\\n\nDATA SET 2:");
    FCFS(data2, n2);

    return 0;
}
```



Tugas 2: Analisis Eksperimen ketiga algoritma

1. Algoritma mana yang memberikan Average Waiting Time terkecil?
- Biasanya SJF memberikan Average Waiting Time (AWT) terkecil.

2. Mengapa hasil algoritma berbeda-beda?
- Karena strategi penjadwalan berbeda (urutan kedatangan, durasi terpendek, atau pembagian quantum).

3. Kapan sebaiknya menggunakan FCFS?
- Sistem batch dengan durasi proses mirip, ketika kesederhanaan diutamakan.

4. Kapan sebaiknya menggunakan SJF?
- Ketika burst time dapat diprediksi, dan tujuan meminimalkan AWT.

5. Kapan sebaiknya menggunakan Round Robin?
- Sistem interaktif / multitasking, butuh respon cepat dan fairness.

Tugas 3: Modifikasi Program

Fitur yang ditambahkan:

- Menampilkan Gantt Chart yang lebih detail
- Menyimpan hasil ke file teks
- Membaca input dari file

```
#include <stdio.h>

struct Process {
    int pid;  // Process ID
    int at;   // Arrival Time
    int bt;   // Burst Time
    int ct;   // Completion Time
    int tat;  // Turnaround Time
    int wt;   // Waiting Time
};

// Fungsi untuk menampilkan hasil dan juga menyimpan ke file
void printAndSaveResult(struct Process p[], int n) {
    FILE *f = fopen("hasil_fcfs.txt", "w");
    if (f == NULL) {
        printf("Gagal membuka file!\n");
        return;
    }

    float avg_tat = 0, avg_wt = 0;

    printf("\n\n=== HASIL PENJADWALAN FCFS ===\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
    fprintf(f, "PID\tAT\tBT\tCT\tTAT\tWT\n");
```

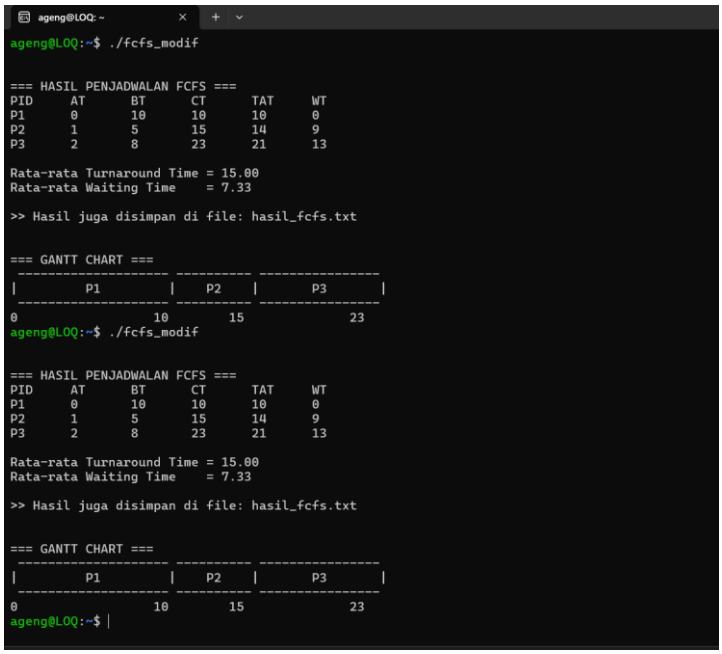
```
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               p[i].pid, p[i].at, p[i].bt, p[i].ct, p[i].tat, p[i].wt);
        fprintf(f, "P%d\t%d\t%d\t%d\t%d\t%d\n",
                p[i].pid, p[i].at, p[i].bt, p[i].ct, p[i].tat, p[i].wt);

        avg_tat += p[i].tat;
        avg_wt += p[i].wt;
    }

    avg_tat /= n;
    avg_wt  /= n;

    printf("\nRata-rata Turnaround Time = %.2f", avg_tat);
    printf("\nRata-rata Waiting Time    = %.2f\n", avg_wt);

    fprintf(f, "\nRata-rata Turnaround Time = %.2f", avg_tat);
    fprintf(f, "\nRata-rata Waiting Time    = %.2f\n", avg_wt);

    fclose(f);
    printf("\n>> Hasil juga disimpan di file: hasil_fcfs.txt\n");
}

// Fungsi untuk menampilkan Gantt Chart
void printGanttChart(struct Process p[], int n) {
    printf("\n\n=== GANTT CHART ===\n");
    printf(" ");

    for (int i = 0; i < n; i++) {
```

```
        for (int j = 0; j < p[i].bt; j++)
            printf("--");
        printf(" ");
    }

    printf("\n|");

    for (int i = 0; i < n; i++) {
        int mid = p[i].bt;
        for (int j = 0; j < mid - 1; j++)
            printf(" ");
        printf("P%d", p[i].pid);
        for (int j = 0; j < mid - 1; j++)
            printf(" ");
        printf("|");
    }

    printf("\n ");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < p[i].bt; j++)
            printf("--");
        printf(" ");
    }

    printf("\n0");
    for (int i = 0; i < n; i++) {
        printf("%*d", p[i].bt * 2, p[i].ct);
    }
    printf("\n");
}
```

```
// Algoritma FCFS
void FCFS(struct Process p[], int n) {
    int time = 0;
    for (int i = 0; i < n; i++) {
        if (time < p[i].at) time = p[i].at;
        time += p[i].bt;
        p[i].ct = time;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;
    }

    printAndSaveResult(p, n);
    printGanttChart(p, n);
}

int main() {
    // Baca dari input.txt (format: pid at bt per baris)
    struct Process p[20];
    int n = 0;

    FILE *f = fopen("input.txt", "r");
    if (f == NULL) {
        printf("File input.txt tidak ditemukan! Menggunakan data default.\n");
        struct Process p_default[] = {
            {1,0,10}, {2,1,5}, {3,2,8}
        };
        n = 3;
        for (int i=0;i<n;i++) p[i] = p_default[i];
    } else {
```

```
        while (fscanf(f, "%d %d %d", &p[n].pid, &p[n].at, &p[n].bt) == 3) n++;
        fclose(f);
    }

    FCFS(p, n);

    return 0;
}
```

Catatan: Buat file input.txt dengan format: pid at bt (satu proses per baris). Contoh:
1 0 10
2 1 5
3 2 8



Tugas 4: Eksperimen dengan data berbeda untuk perbandingan algoritma

Test Case 1: Proses dengan burst time bervariasi

Jumlah proses: 4

Time quantum: 3

- P1: AT=0, BT=24
- P2: AT=1, BT=3
- P3: AT=2, BT=3
- P4: AT=3, BT=6

Test Case 2: Proses datang bersamaan

Jumlah proses: 5

Time quantum: 4

- P1: AT=0, BT=10
- P2: AT=0, BT=5
- P3: AT=0, BT=8
- P4: AT=0, BT=12
- P5: AT=0, BT=6

Test Case 3: Proses datang dengan interval

Jumlah proses: 5

Time quantum: 2

- P1: AT=0, BT=8
- P2: AT=2, BT=4
- P3: AT=4, BT=9
- P4: AT=6, BT=5
- P5: AT=8, BT=3

CODE

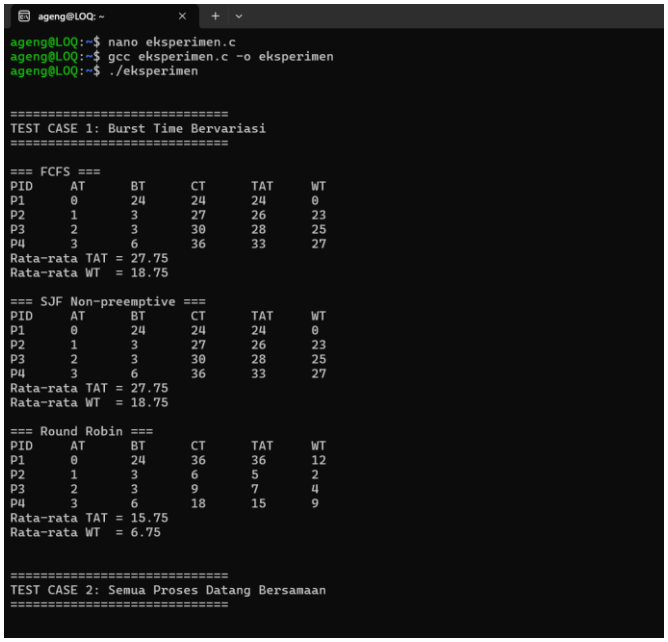
```
#include <stdio.h>

struct Process {
    int pid;
    int at;
    int bt;
    int ct;
    int tat;
    int wt;
    int rt; // remaining time (untuk RR)
};

// Cetak hasil
void printResult(struct Process p[], int n, char *algoname) {
    float avg_tat = 0, avg_wt = 0;
    printf("\n=== %s ===\n", algoname);
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
    for (int i = 0; i < n; i++) {
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;
        avg_tat += p[i].tat;
        avg_wt += p[i].wt;
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
               p[i].pid, p[i].at, p[i].bt, p[i].ct, p[i].tat, p[i].wt);
    }
    printf("Rata-rata TAT = %.2f\n", avg_tat / n);
```

```
    printf("Rata-rata WT  = %.2f\n", avg_wt / n);
}

// FCFS
void FCFS(struct Process p[], int n) {
    int time = 0;
    for (int i = 0; i < n; i++) {
        if (time < p[i].at) time = p[i].at;
        time += p[i].bt;
        p[i].ct = time;
    }
    printResult(p, n, "FCFS");
}

// SJF Non-preemptive
void SJF(struct Process p[], int n) {
    int completed = 0, time = 0;
    int done[10] = {0};

    while (completed < n) {
        int idx = -1, min_bt = 9999;
        for (int i = 0; i < n; i++) {
            if (!done[i] && p[i].at <= time && p[i].bt < min_bt) {
                min_bt = p[i].bt;
                idx = i;
            }
        }

        if (idx == -1) {
            time++;
```

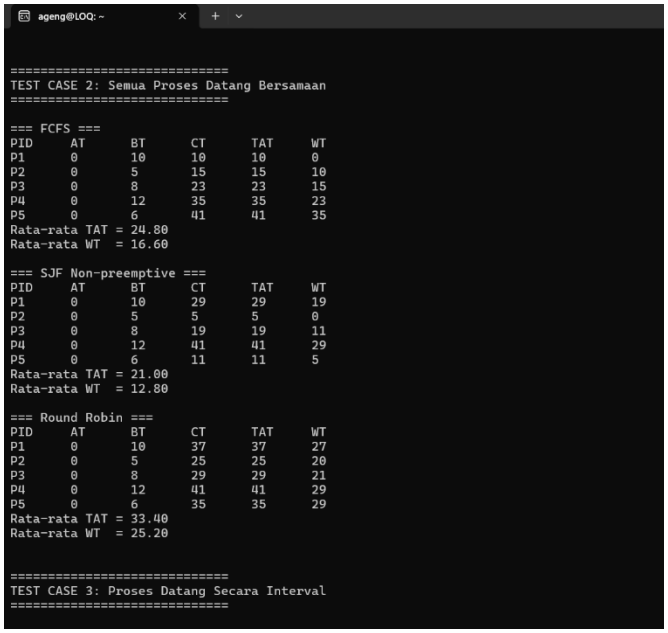
```
            continue;
        }

        time += p[idx].bt;
        p[idx].ct = time;
        done[idx] = 1;
        completed++;
    }
    printResult(p, n, "SJF Non-preemptive");
}

// Round Robin
void RR(struct Process p[], int n, int tq) {
    int completed = 0;
    int time = 0;

    for (int i = 0; i < n; i++)
        p[i].rt = p[i].bt; // remaining time

    while (completed < n) {
        int all_done = 1;
        for (int i = 0; i < n; i++) {
            if (p[i].rt > 0 && p[i].at <= time) {
                all_done = 0;
                if (p[i].rt > tq) {
                    time += tq;
                    p[i].rt -= tq;
                } else {
                    time += p[i].rt;
                    p[i].rt = 0;
```

```
                p[i].ct = time;
                completed++;
            }
        } else if (p[i].rt > 0 && p[i].at > time) {
            time++; // tunggu proses berikutnya datang
        }
    }
    if (all_done) break;
  }
  printResult(p, n, "Round Robin");
}

// Salin array agar tidak tercampur antar algoritma
void copyData(struct Process src[], struct Process dest[], int n) {
  for (int i = 0; i < n; i++)
    dest[i] = src[i];
}

// Jalankan satu test case
void runTestCase(struct Process data[], int n, int tq, char *judul) {

printf("\n\n==============================\n%s\n==========================
======\n", judul);

  struct Process fcfs[10], sjf[10], rr[10];
  copyData(data, fcfs, n);
  copyData(data, sjf, n);
  copyData(data, rr, n);

  FCFS(fcfs, n);
  SJF(sjf, n);
```

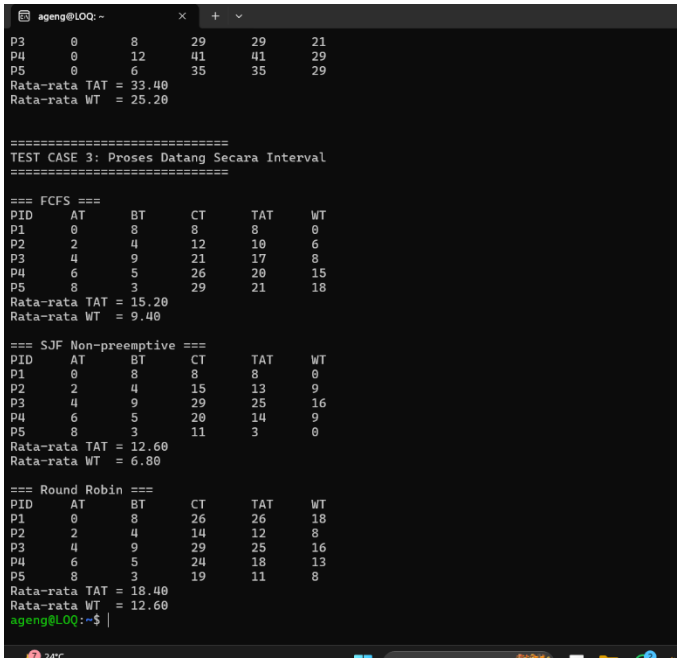
```
    RR(rr, n, tq);
}

int main() {
    // TEST CASE 1
    struct Process tc1[] = {
        {1, 0, 24}, {2, 1, 3}, {3, 2, 3}, {4, 3, 6}
    };
    runTestCase(tc1, 4, 3, "TEST CASE 1: Burst Time Bervariasi");

    // TEST CASE 2
    struct Process tc2[] = {
        {1, 0, 10}, {2, 0, 5}, {3, 0, 8}, {4, 0, 12}, {5, 0, 6}
    };
    runTestCase(tc2, 5, 4, "TEST CASE 2: Semua Proses Datang Bersamaan");

    // TEST CASE 3
    struct Process tc3[] = {
        {1, 0, 8}, {2, 2, 4}, {3, 4, 9}, {4, 6, 5}, {5, 8, 3}
    };
    runTestCase(tc3, 5, 2, "TEST CASE 3: Proses Datang Secara Interval");

    return 0;
}
```



```
ageng@LOQ: ~
ageng@LOQ:~$ nano eksperimen.c
ageng@LOQ:~$ gcc eksperimen.c -o eksperimen
ageng@LOQ:~$ ./eksperimen


============================
TEST CASE 1: Burst Time Bervariasi
============================

=== FCFS ===
PID     AT      BT      CT      TAT     WT
P1      0       24      24      24      0
P2      1       3       27      26      23
P3      2       3       30      28      25
P4      3       6       36      33      27
Rata-rata TAT = 27.75
Rata-rata WT  = 18.75

=== SJF Non-preemptive ===
PID     AT      BT      CT      TAT     WT
P1      0       24      24      24      0
P2      1       3       27      26      23
P3      2       3       30      28      25
P4      3       6       36      33      27
Rata-rata TAT = 27.75
Rata-rata WT  = 18.75

=== Round Robin ===
PID     AT      BT      CT      TAT     WT
P1      0       24      36      36      12
P2      1       3       6       5       2
P3      2       3       9       7       4
P4      3       6       18      15      9
Rata-rata TAT = 15.75
Rata-rata WT  = 6.75


============================
TEST CASE 2: Semua Proses Datang Bersamaan
============================
```



```
ageng@LOQ: ~


============================
TEST CASE 2: Semua Proses Datang Bersamaan
============================

=== FCFS ===
PID     AT      BT      CT      TAT     WT
P1      0       10      10      10      0
P2      0       5       15      15      10
P3      0       8       23      23      15
P4      0       12      35      35      23
P5      0       6       41      41      35
Rata-rata TAT = 24.80
Rata-rata WT  = 16.60

=== SJF Non-preemptive ===
PID     AT      BT      CT      TAT     WT
P1      0       10      29      29      19
P2      0       5       5       5       0
P3      0       8       19      19      11
P4      0       12      41      41      29
P5      0       6       11      11      5
Rata-rata TAT = 21.00
Rata-rata WT  = 12.80

=== Round Robin ===
PID     AT      BT      CT      TAT     WT
P1      0       10      37      37      27
P2      0       5       25      25      20
P3      0       8       29      29      21
P4      0       12      41      41      29
P5      0       6       35      35      29
Rata-rata TAT = 33.40
Rata-rata WT  = 25.20


============================
TEST CASE 3: Proses Datang Secara Interval
============================
```


ageng@LOQ: ~
P3 0 8 29 29 21
P4 0 12 41 41 29
P5 0 6 35 35 29
Rata-rata TAT = 33.40
Rata-rata WT = 25.20
==========
TEST CASE 3: Proses Datang Secara Interval
==========
=== FCFS ===
PID AT BT CT TAT WT
P1 0 8 8 8 0
P2 2 4 12 10 6
P3 4 9 21 17 8
P4 6 5 26 20 15
P5 8 3 29 21 18
Rata-rata TAT = 15.20
Rata-rata WT = 9.40
=== SJF Non-preemptive ===
PID AT BT CT TAT WT
P1 0 8 8 8 0
P2 2 4 15 13 9
P3 4 9 29 25 16
P4 6 5 20 14 9
P5 8 3 11 3 0
Rata-rata TAT = 12.60
Rata-rata WT = 6.80
=== Round Robin ===
PID AT BT CT TAT WT
P1 0 8 26 26 18
P2 2 4 14 12 8
P3 4 9 29 25 16
P4 6 5 24 18 13
P5 8 3 19 11 8
Rata-rata TAT = 18.40
Rata-rata WT = 12.60
ageng@LOQ:~$

Tugas 5: Analisis Eksperimen Perbandingan Algoritma

1. Analisis Performa
- FCFS terbaik ketika proses memiliki burst time mirip dan sistem batch.
- SJF hampir selalu memberikan Average WT terkecil karena memilih job terpendek lebih dulu (optimal untuk AWT jika durasi diketahui).
- Trade-off RR: adil dan responsif tetapi banyak context switch jika quantum kecil.

2. Context Switching
- RR paling banyak melakukan context switch.
- Quantum kecil -> banyak context switch; quantum besar -> sedikit context switch.

3. Fairness
- RR paling adil.
- FCFS merugikan proses pendek jika ada proses panjang di depan (convoy effect).
- SJF berisiko starvation untuk proses panjang.

4. Real World Application
- Web server (many small requests): SJF atau RR dengan quantum kecil.
- Render video (task besar): FCFS atau SJF.
- OS desktop (interactive): Round Robin.